Jiani (Elva) He

Data Science Consultant | Independent Researcher
Seattle, Washington | jiani19980225@gmail.com
jiani19980225.github.io | LinkedIn | Google Scholar

PROFESSIONAL EXPERIENCE

Accenture | Data Science Consultant

Seattle, WA | Jun 2024 - Present

- Lead planning analyses and client relationships; present recommendations to senior leaders.
- Analyze reverse-logistics cycle time and on-time delivery using SQL; support reuse, resale, recycling, and disposal decisions based on asset value and lifecycle stage.
- Analyze demand, stock targets, and inventory; use optimization to rebalance supply across sites.
- Plan quality-assurance component quantities using demand and bill-of-materials data; define service metrics and support inventory adjustments.
- Develop reusable prompt and query guides for enterprise LLM assistants; train analysts in AI-assisted analysis.

Accenture | Data Science Senior Analyst

Seattle, WA | Oct 2021 - May 2024

- Built SQL dashboards and shortage alerts to compare inventory with demand and inform site transfers.
- Tracked reverse-logistics cycle times and delivery performance to support asset recovery scheduling.
- Investigated inventory imbalances and abnormal cycle times with statistical analysis; automated reporting and purchase-order workflows.
- Across both Accenture roles, mentored five junior analysts and led onboarding and team training.

BMW Group | Data Scientist Intern, MIT Capstone

Cambridge, MA | Feb - Aug 2021

- Modeled vehicle demand and cross-price elasticity in Python to recommend discounts by customer segment and sales channel; built Tableau pricing dashboards.

EDUCATION

Massachusetts Institute of Technology

MS, Business Analytics | Operations Research Center | Aug 2021 | GPA: 4.7/5.0

Classification trees for 40,000 news articles; robust bike allocation in Julia; NLP admissions Q&A pipeline for AdmitHub, including response evaluation and human review.

Purdue University

BS, Industrial Engineering | May 2020 | Honors & Distinction | GPA: 3.9/4.0

LSTM forecasting in Keras and reinforcement learning for sensor placement in Python/R.

Research, skills & service

Jiani (Elva) He | Curriculum vitae | Updated October 2026

RESEARCH & PREPRINTS

Semantic Signal-Assisted Inspection and Recovery Allocation in Reverse Logistics

Jiani He, Dingyan Shang, Yihua Xu, Shiqi Huang, Yan Lyu, Jize Li, Shangjing Tang. 2026.

Accepted · AIBThings 2026. Public preprint · arXiv:2609.02116.

https://arxiv.org/abs/2609.02116

When Does an Added Planning Capability Create Operational Value? An Observable Evidence Standard

Yihua Xu, Yan Lyu, Jize Li, **Jiani He**, You Lin, Yidan Hu, Rubing Qiang. 2026.

Accepted & presented · IEOM Irvine 2026. Public preprint · arXiv:2610.27305.

https://arxiv.org/abs/2610.27305

Policy-Bounded, Provenance-Aware Deployment Assurance for LLM Request Routing

Yihua Xu, **Jiani He**, Ishita Chirag Talati, Yitian Qian, Youting Wang, Zhenyu Xu, Dingyan Shang. 2026.

Preprint · SSRN. SSRN abstract 7554440.

https://papers.ssrn.com/sol3/papers.cfm?abstract_id=7554440

DACRI: Decision-Aware Causal Intervention Ranking for Critical Supply Chains

Shiqi Huang, **Jiani He**, Dingyan Shang, Yihua Xu, Jize Li, Yan Lyu, Lashimi Muraleedharan Nair. 2026.

Accepted for presentation · ICECCME 2026. Public preprint · arXiv:2608.11154.

https://arxiv.org/abs/2608.11154

Accuracy-Preserving Stability Regularization for Large-Scale Retail Demand Forecasting

Jize Li, **Jiani He**, Dishu Yang, Dingyan Shang, Jingjing Liu, Shiqi Huang. 2026.

Accepted for presentation · ICEME 2026. Public preprint · arXiv:2607.13331.

https://arxiv.org/abs/2607.13331

TECHNICAL & CONSULTING SKILLS

Tools: SQL, Python, R, Julia, Keras, Tableau, AWS

Modeling: Machine learning, forecasting, regression, classification, clustering, robust optimization

Analytics & AI: Statistical analysis, root-cause analysis, visualization, dashboards, KPI design, NLP, generative AI, LLM workflows, prompt and query design

Consulting: Problem framing, client communication, stakeholder coordination, project delivery, mentoring

SERVICE & MENTORING

Peer-reviewed 10+ journal manuscripts. Mentored five junior analysts across Accenture roles in SQL, business problem solving, and client communication; led onboarding and team training.